EART 22201 Igneous minerals and processes

Volcanic eruptions



Lesson 10

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Volcanic eruptions

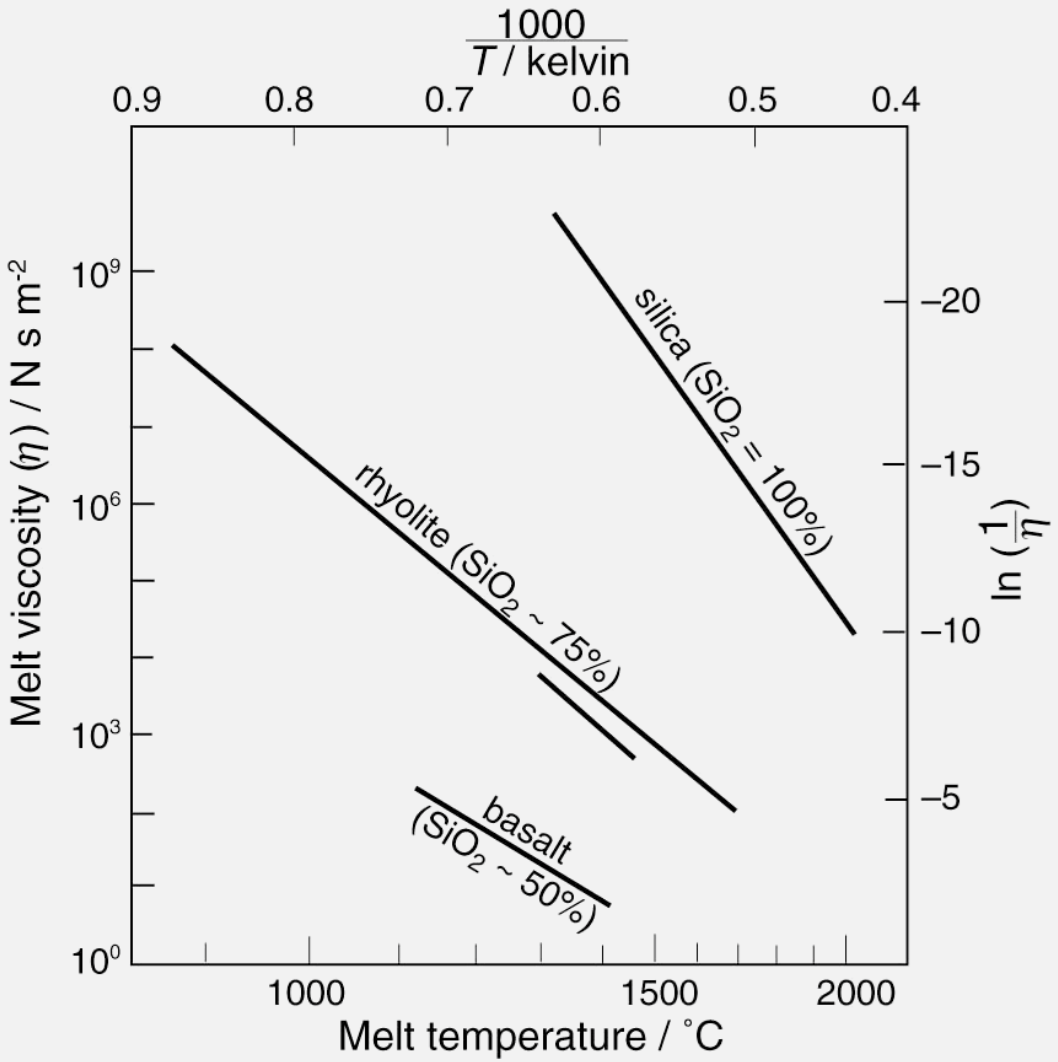
1.1 Viscosity, volatiles, and volcanic degassing

1.2 What happens during magma ascent?



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Viscosity controls how easily a magma can flow



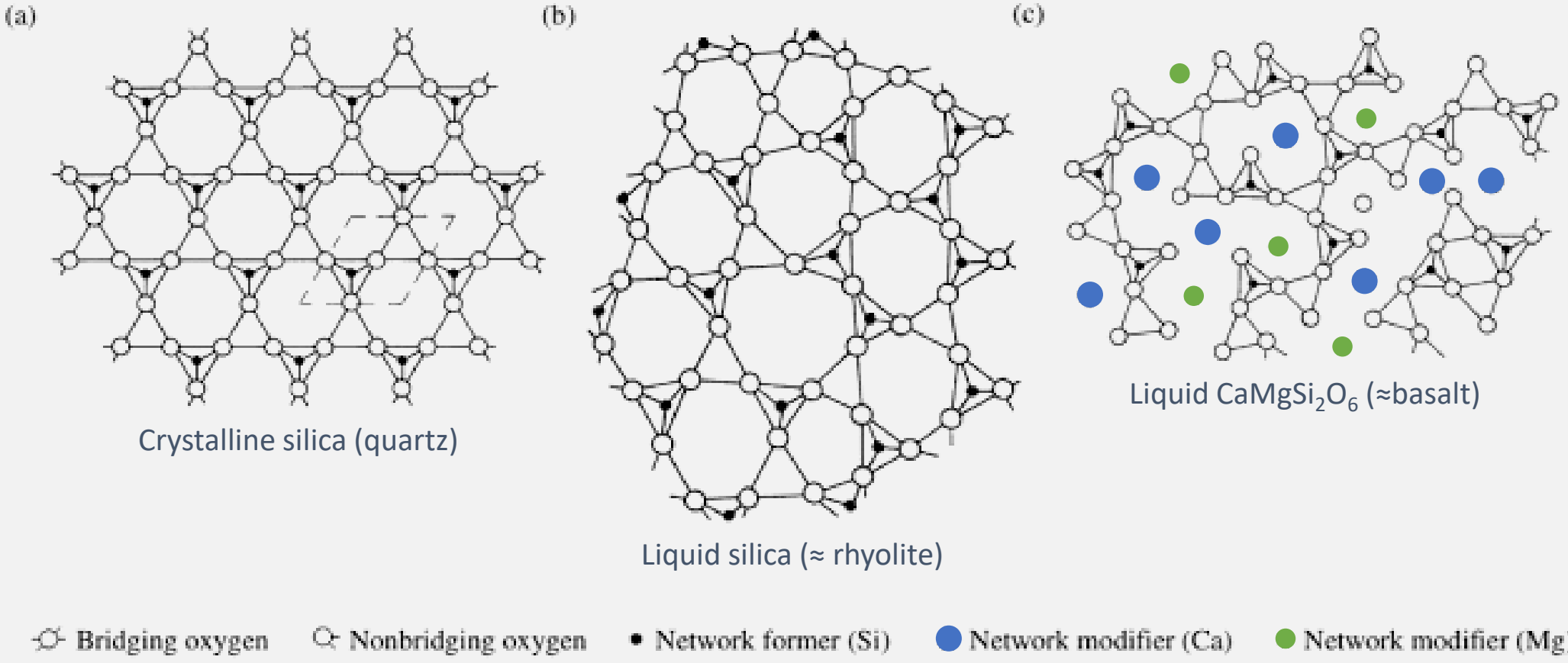
$$\tau = \eta\left(\frac{du}{dy}\right)$$

η = viscosity	Pa.s
τ = shear stress	Pa
$\left(\frac{du}{dy}\right)$ = shear rate	s^{-1}

At a fixed melt SiO_2 content, viscosity is strongly controlled by temperature.

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Viscosity is strongly controlled by melt SiO_2 content



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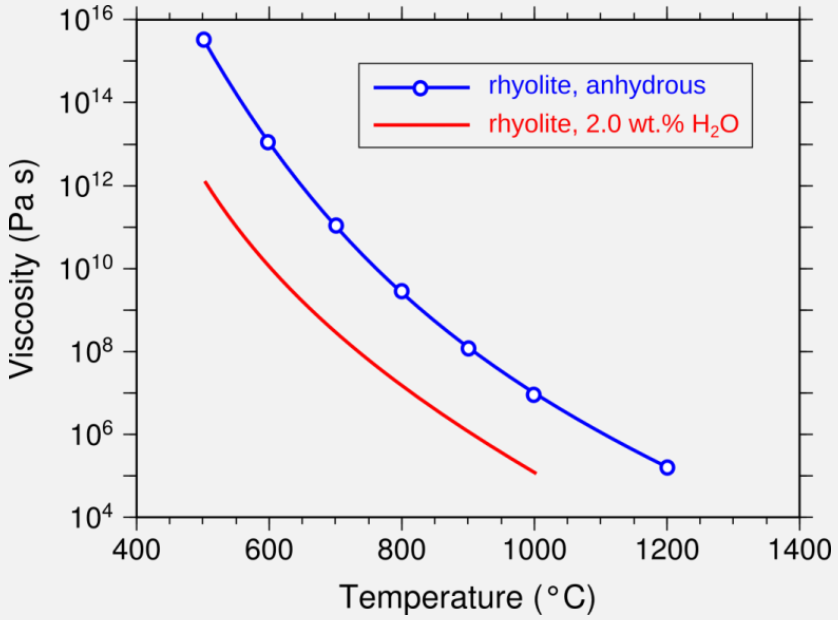
Viscosity is strongly controlled by melt H_2O content



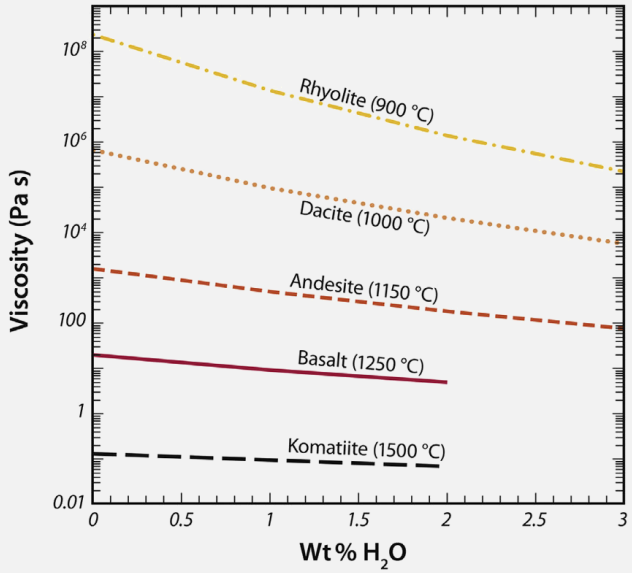
Si-O polymer in anhydrous melt

water molecule

broken Si-O polymer in hydrous melt

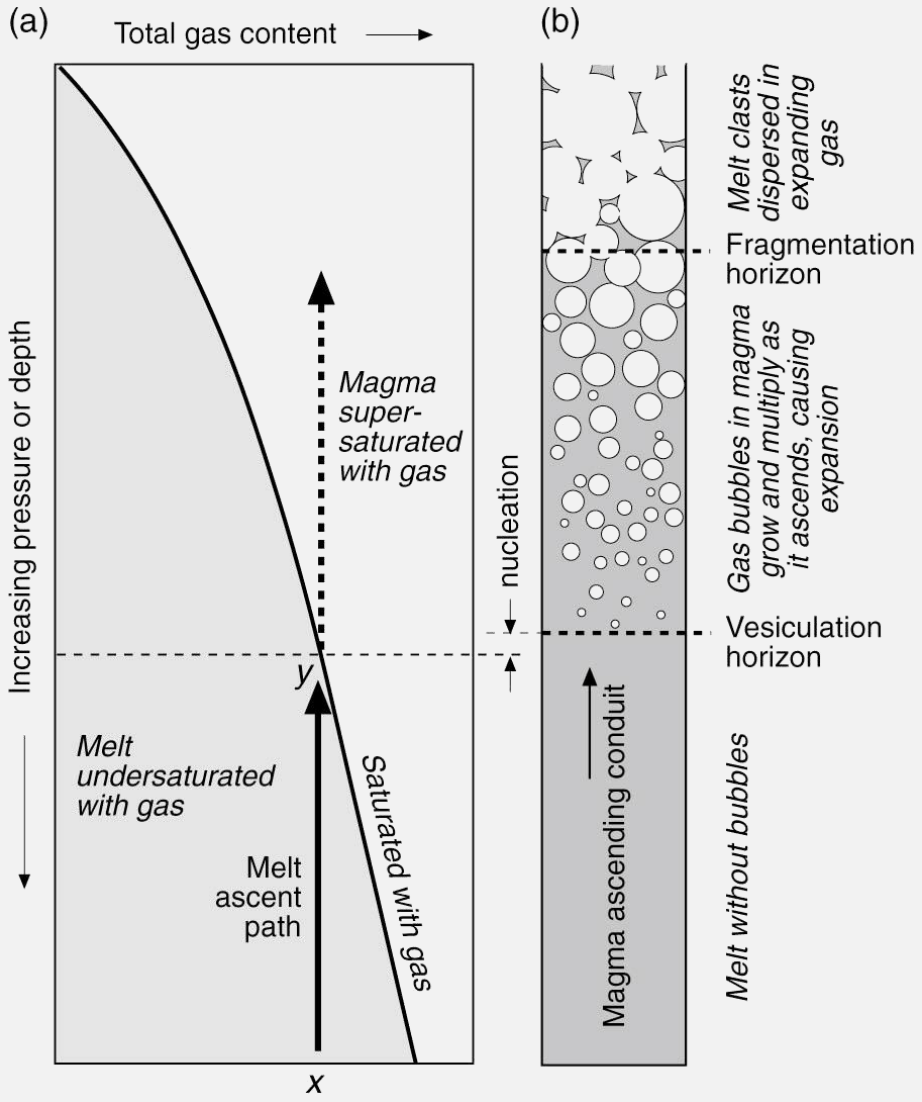


- At fixed melt SiO_2 content, adding H_2O decreases the viscosity
- The higher the melt SiO_2 content, the more H_2O can be dissolved into the melt
- When H_2O saturation is reached, no more water can be dissolved into the melt



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Volatile solubility and degassing



Fragmentation: If viscosity and bubble pressure are high enough, an explosive eruption will occur.

Vesiculation: H_2O exsolves and gas bubbles form. At the same time, T is decreasing and the crystal content is increasing, so viscosity increases.

High pressure: All H_2O is entirely dissolved in the melt.

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Magma viscosity controls vesiculation behaviour and vesicle textures

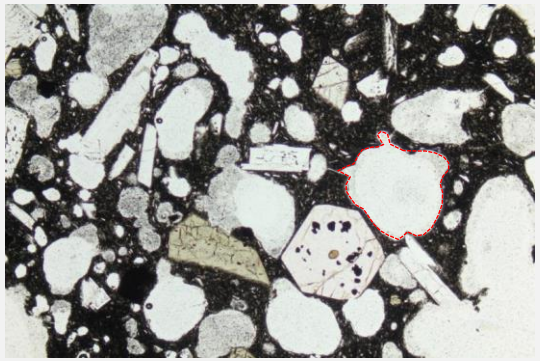
Basaltic scoria



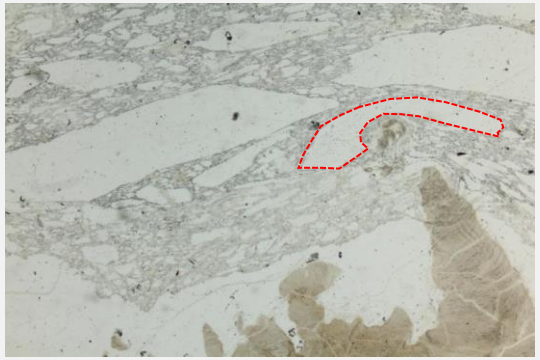
Rhyolitic pumice



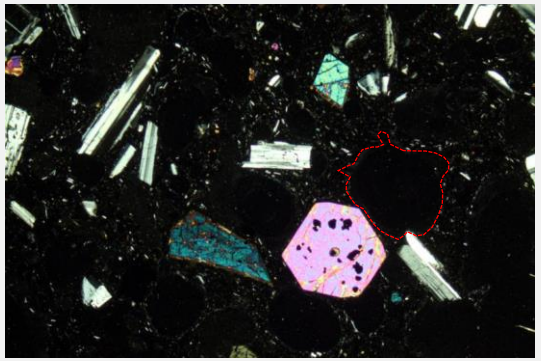
plane-polarized light



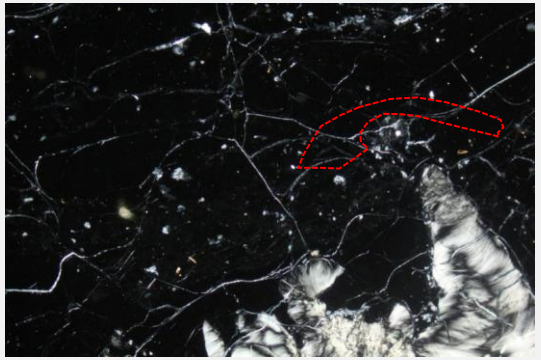
In PPL, vesicles are colourless, transparent, featureless



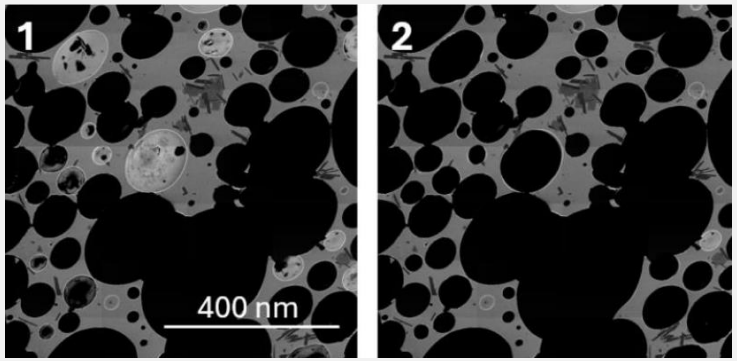
crossed polars



In XP, vesicles are always in extinction

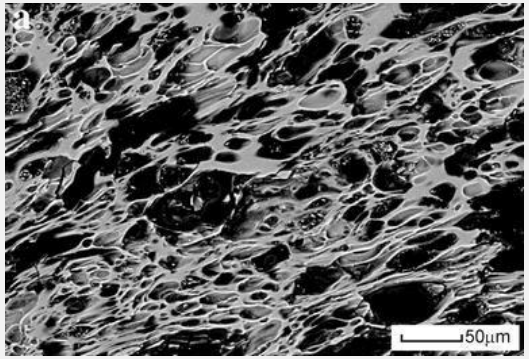


scanning electron microscope



Basalt: vesicles tend to be rounded

Rhyolite: vesicles tend to be elongate



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Volcanic eruptions

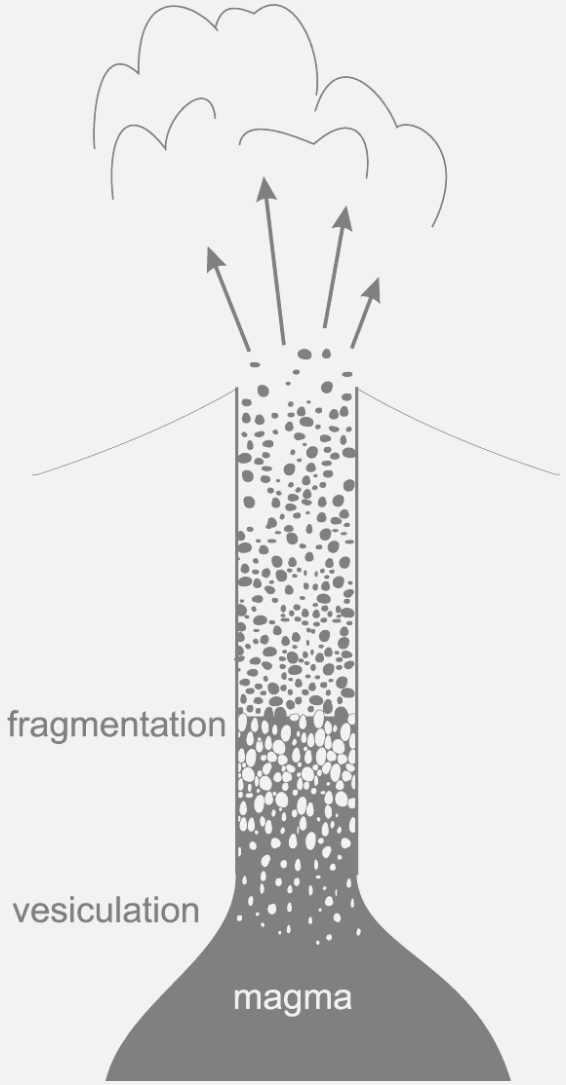
1.1 Viscosity, volatiles, and volcanic degassing

1.2 What happens during magma ascent?



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Evolution of magma viscosity during ascent through a volcanic conduit



In conduit, during magma ascent (decreasing pressure):

- Temperature decreasing
- H_2O solubility decreasing
- Crystal content increasing

Viscosity increases

In magma reservoir at depth (high pressure):

- Temperature is high*
- H_2O solubility is high*
- Crystal content is low*

Viscosity is low*

*compared to the erupted magma

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Volcanic eruptions

You should now be able to:

- Describe how and why magma viscosity varies as a function of temperature, melt composition and water content
- Describe the processes of degassing, vesiculation and fragmentation
- Explain the feedbacks between viscosity, temperature, volatile content, and crystal content, and relate these factors to the likelihood of an explosive eruption occurring



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Volcanic eruptions

2.1 Effusive eruptions and their products



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Effusive eruptions: Basalt lavas

pāhoehoe

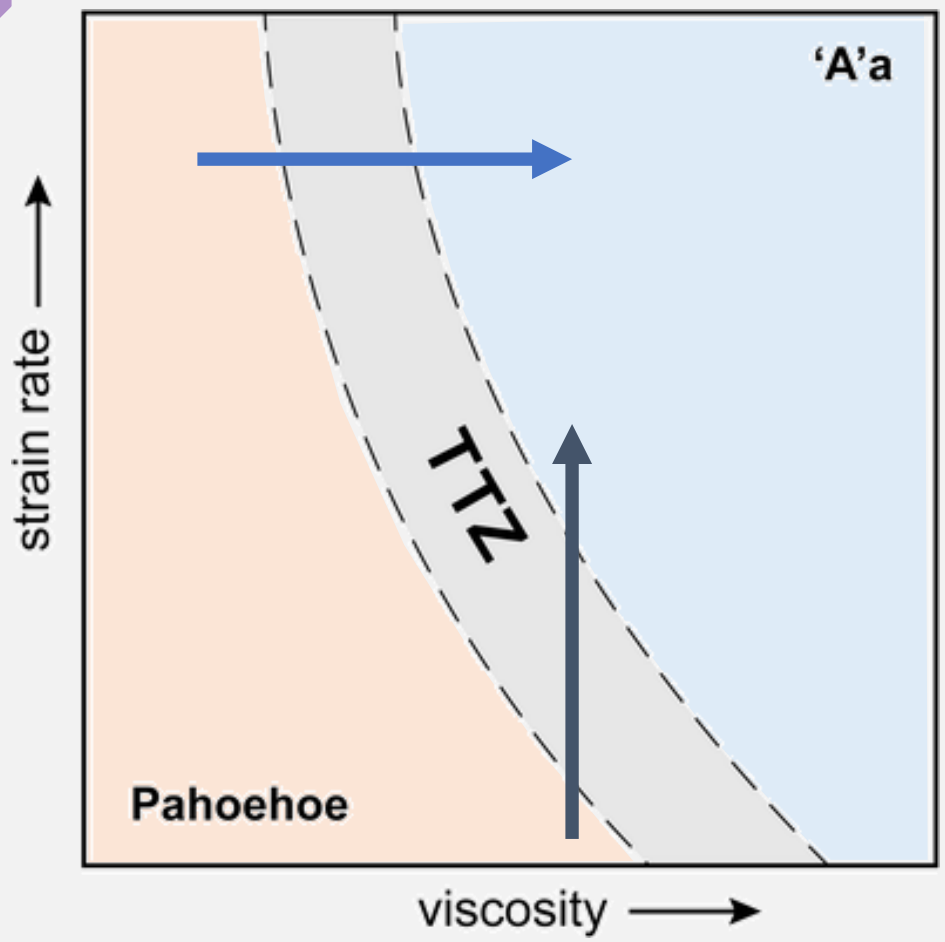


ʻaʻā



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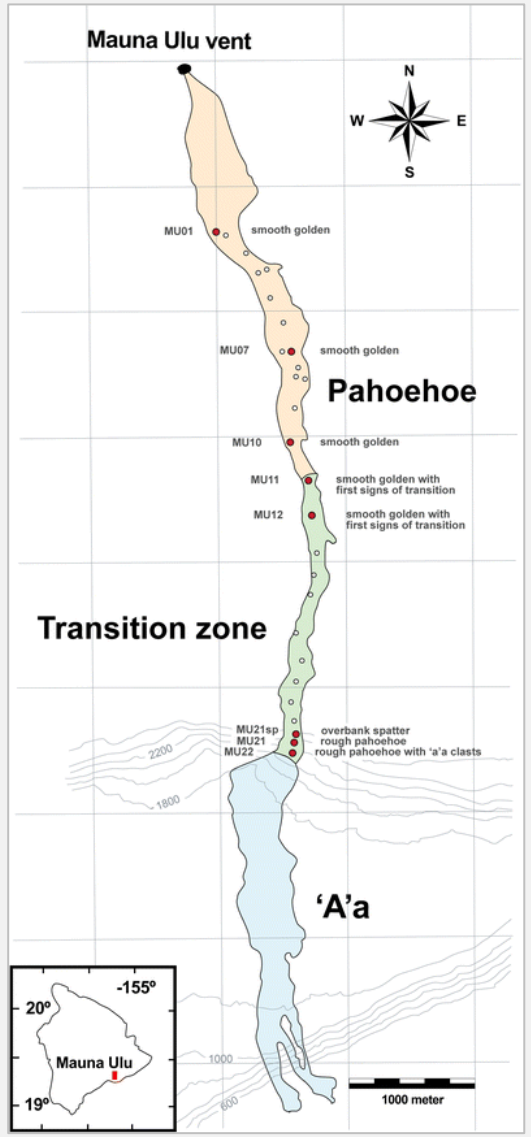
Effusive eruptions: Basalt lavas



Increasing viscosity at constant shear rate by cooling and crystallization

Increasing shear rate at constant viscosity, e.g. lava flows over a steeper slope

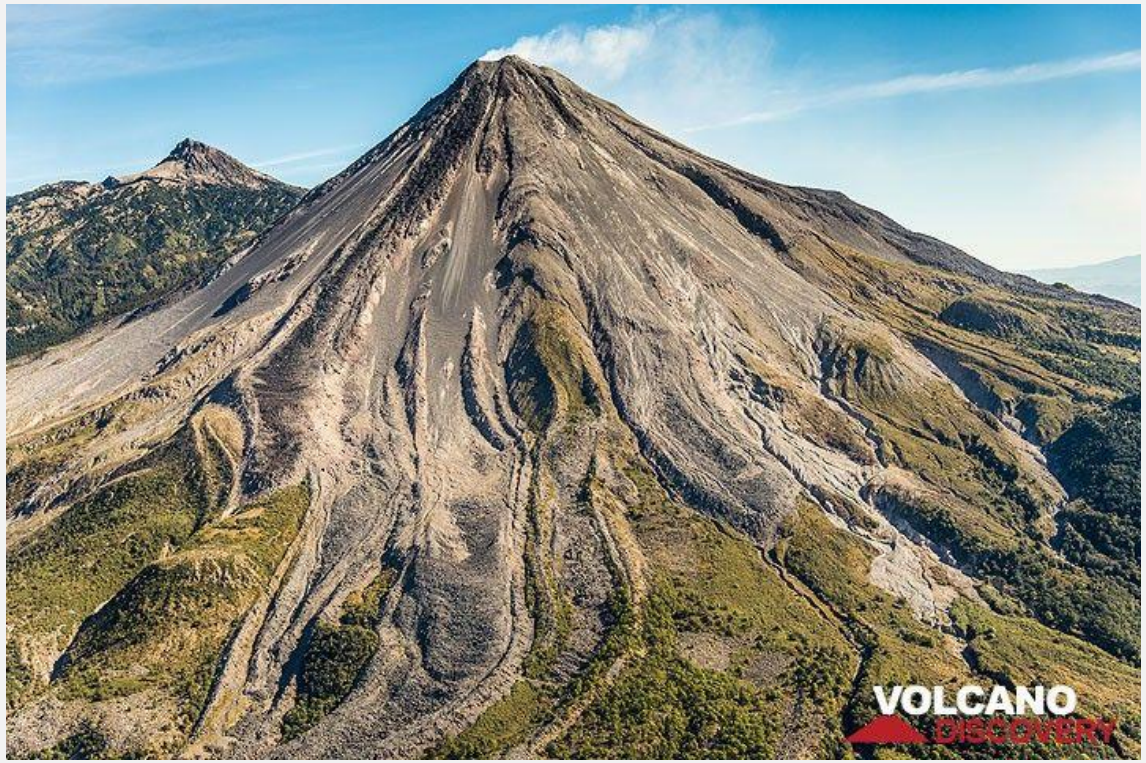
TTZ = transition threshold zone



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Effusive eruptions: Andesite lavas

Colima, Mexico



Kanaga, Aleutian Islands



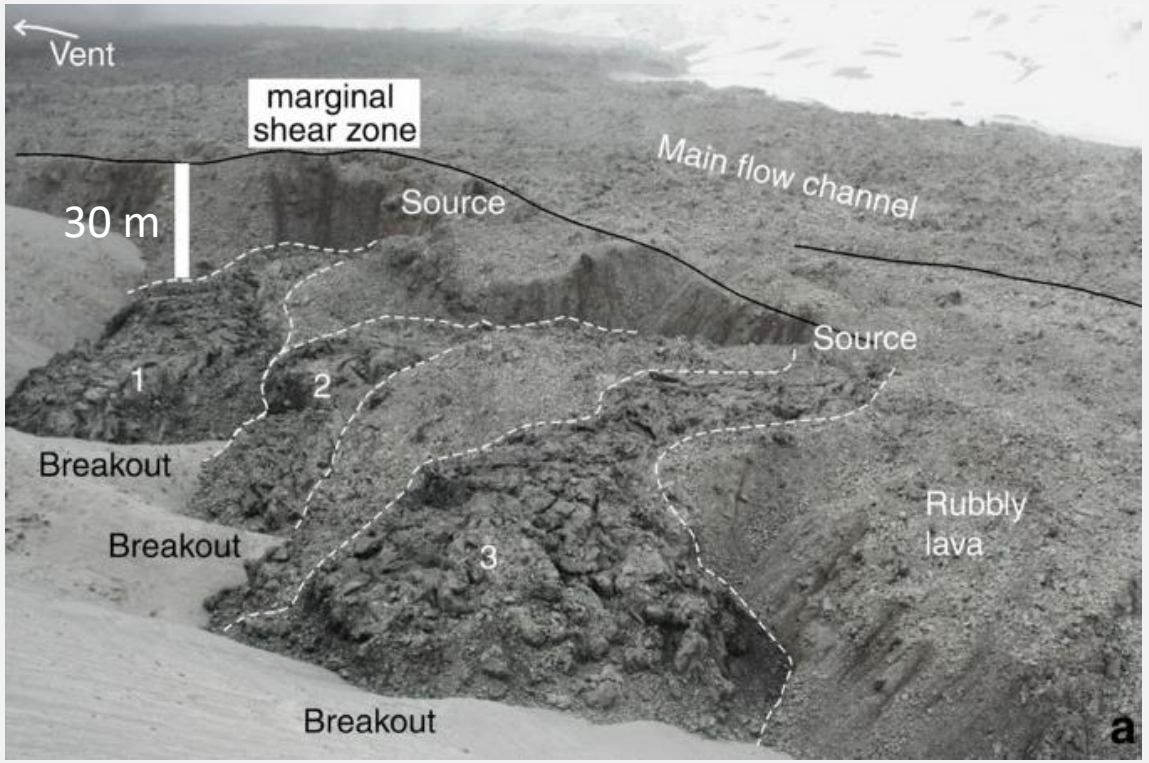
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Effusive eruptions: Rhyolite lavas

Landmannalaugar, Iceland



Puyehue-Cordón Caulle, Chile



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Effusive eruptions: Lava domes and spines



Dacitic lava domes at La Soufrière, St Vincent:

2020-2021 (fresh, black)

1971-1972 (vegetated, green)

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Effusive eruptions: Lava domes and spines



Mount St Helens, 2006

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Volcanic eruptions

You should now be able to:

- Qualitatively describe the products of effusive activity
- Relate the morphology of effusive eruption products to magma composition and viscosity



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Volcanic eruptions

3.1 Explosive eruption styles and the Volcanic Explosivity Index

3.2 Products of explosive eruptions

- Pyroclastic fall deposits
- Pyroclastic density current deposits

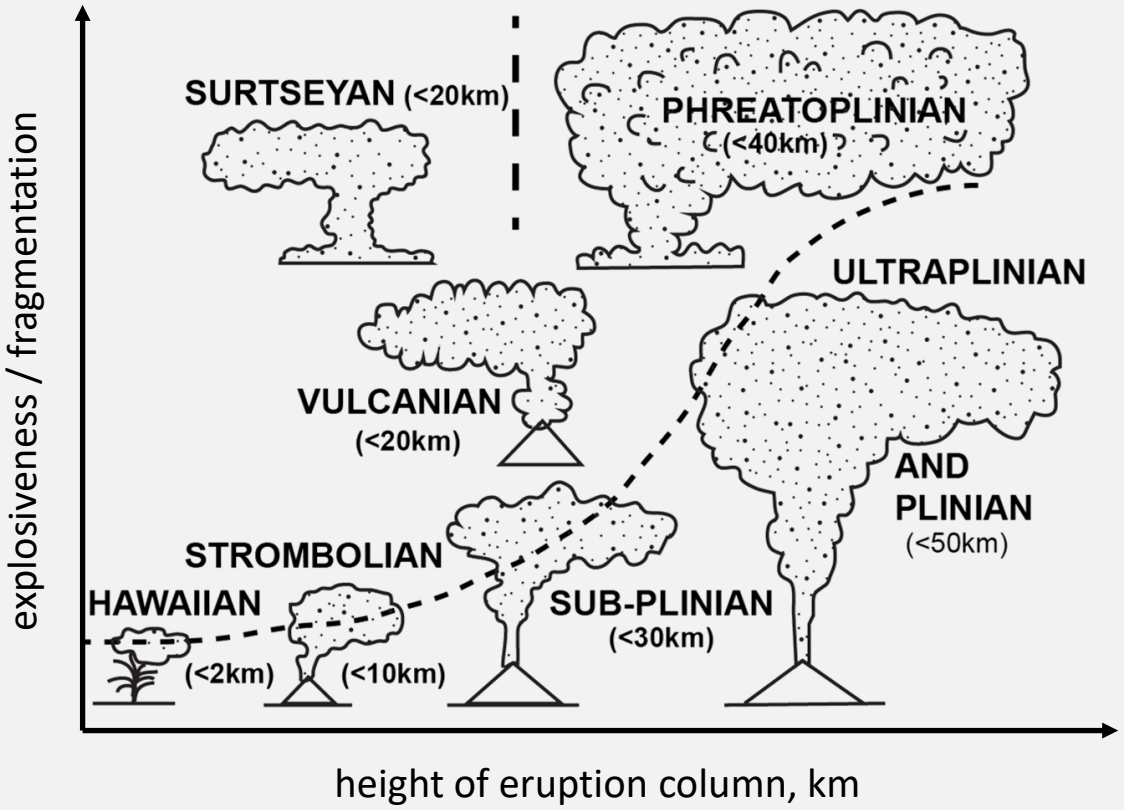
3.3 Reconstructing ancient eruptions from their deposits

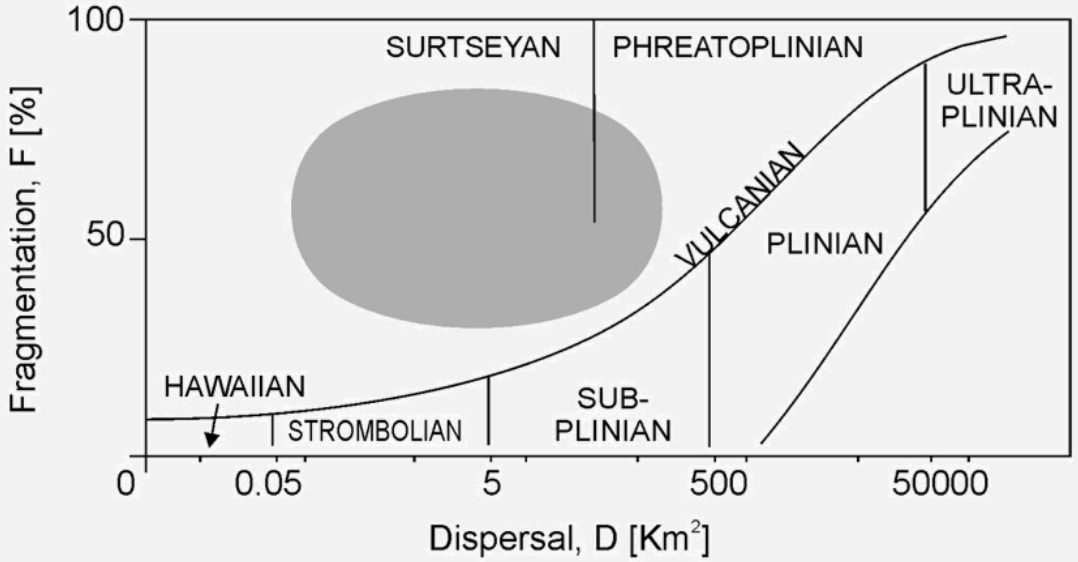
3.4 Calderas



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Explosive eruptions





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Explosive eruption styles: Hawaiian

Mauna Ulu, 1969



Fissure 8, Leilani Estates eruption, 2018



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Explosive eruption styles: Strombolian

Anak Krakatau, Indonesia, 2008



Stromboli, Italy, 2019



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Explosive eruption styles: Vulcanian

Sakurajima, Japan, 2009



Stromboli paroxysmal eruption, 2019



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Explosive eruption styles: Plinian

Redoubt, Alaska, 1990



Calbuco, Chile, 2015



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Some type examples of explosive eruption styles

YouTube video footage

Hawaiian: Drone footage of Fissure 8, Leilani Estates eruption, Kilauea, Hawaii, 2018

Strombolian: Dual crater Strombolian explosions, Fogo Volcano, Cabo Verde, 2014

Vulcanian: Eruption and small pyroclastic flow, Sakurajima, Japan, 2013

Sub-Plinian: Eruption column and ash fall, Etna, Italy, 2021

Volcano news and updates

The **Global Volcanism Program** releases a weekly bulletin describing new and ongoing volcanic activity and unrest.

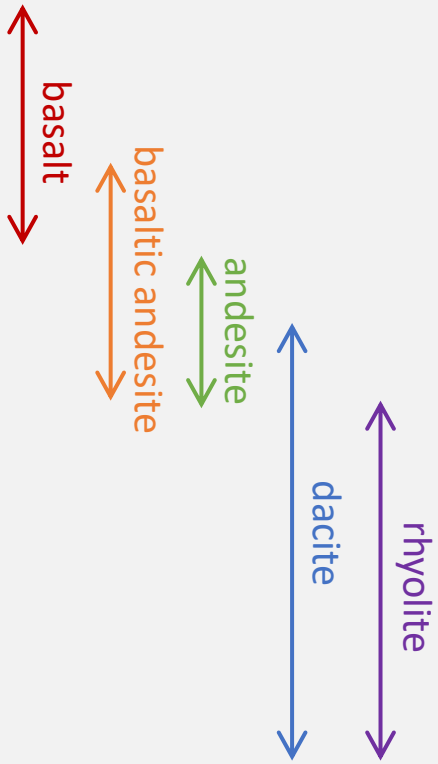
It also has an excellent database describing past activity and typical eruption styles for most global volcanoes.

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The Volcanic Explosivity Index: a quantitative approach to eruption explosivity

VEI	Description	Plume height	Volume	Classification	Frequency	Example
0	non-explosive	<100 m	<10,000 m^3	Hawaiian	constant/daily	Kilauea
1	gentle	0.1-1 km	>10,000 m^3	Hawaiian/Strombolian	daily	Stromboli
2	explosive	1-5 km	>1,000,000 m^3	Strombolian/Vulcanian	weekly	Whakaari 2019
3	severe	3-15 km	>0.01 km^3	Vulcanian	few months	Anak Krakatau 2018
4	cataclysmic	10-25 km	>0.1 km^3	Vulcanian/Plinian	≥1 yr	Eyjafjallajökull 2010
5	paroxysmal	>25 km	>1 km^3	Plinian	≥10 yr	Mount St Helens 1980
6	colossal	>25 km	>10 km^3	Plinian/ultra-Plinian	≥100 yr	Krakatau 1883
7	super-colossal	>25 km	>100 km^3	ultra-Plinian	≥1000 yr	Tambora 1815
8	mega-colosal	>25 km	>1000 km^3	ultra-Plinian	≥10,000 yr	Yellowstone 2 Ma

*There are rare examples of basaltic Plinian eruptions in the geological record



Pyle DM (2000) Sizes of volcanic eruptions. Encyclopedia of Volcanoes, pp 263-269

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Volcanic eruptions

You should now be able to:

- Qualitatively describe styles of explosive activity
- Relate the style of explosive activity to magma composition and viscosity
- Explain the concept of the Volcanic Explosivity Index, and list the parameters used to quantify eruption explosivity



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Volcanic eruptions

3.1 Explosive eruption styles and the Volcanic Explosivity Index

3.2 Products of explosive eruptions

- Pyroclastic fall deposits

- Pyroclastic density current deposits

3.3 Reconstructing ancient eruptions from their deposits

3.4 Calderas

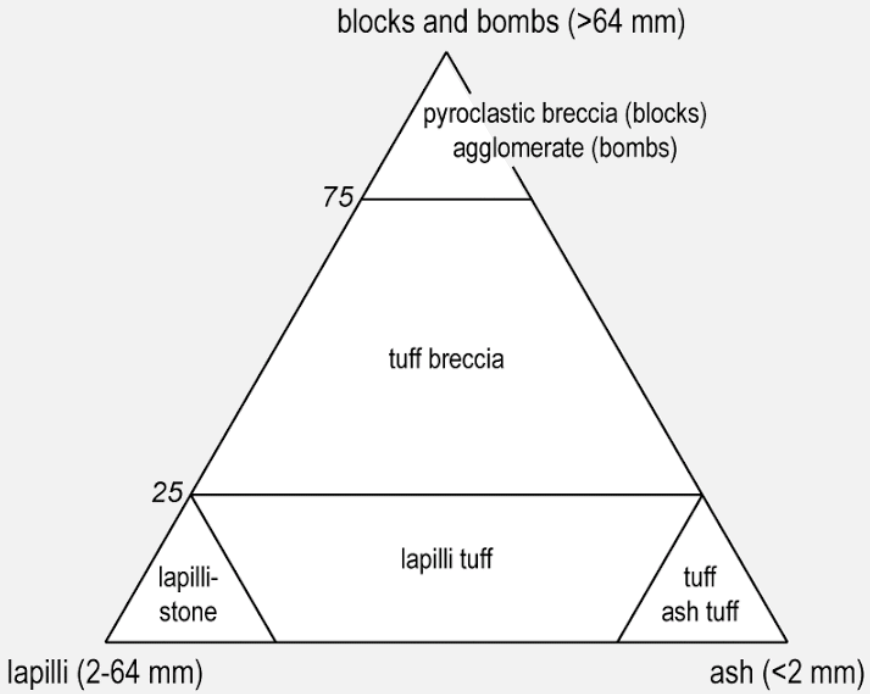


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Pyroclasts from explosive eruptions

Tephra: the fragmental material produced by a volcanic eruption – regardless of composition, fragment size, or emplacement mechanism.

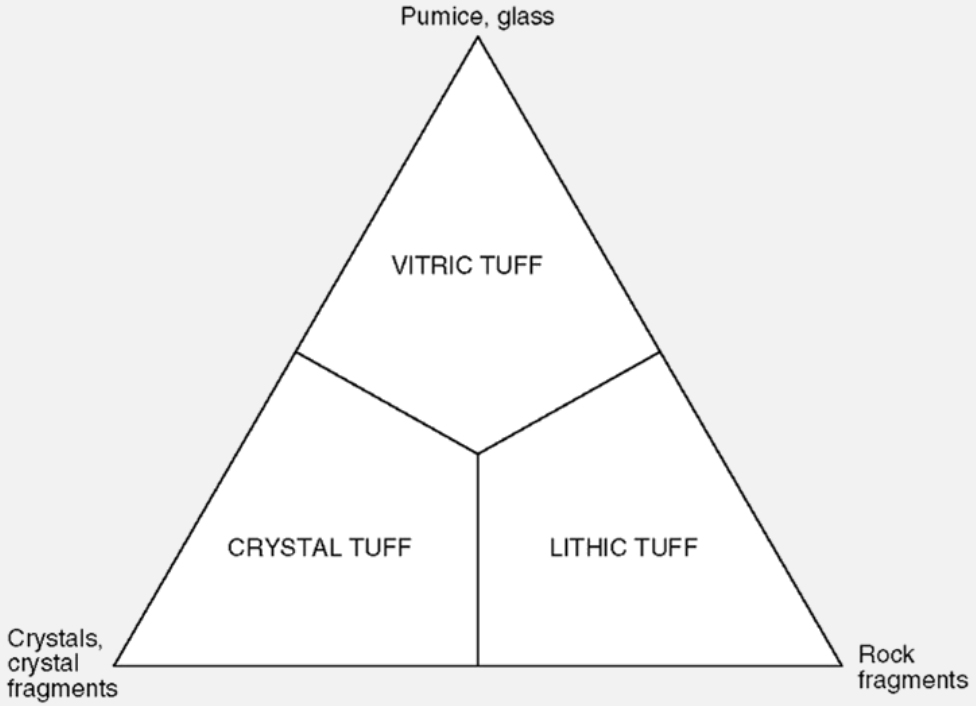
Grain size, mm	Volcanic clast type	Sedimentary equivalent
>256	coarse blocks and bombs	boulders
64-256	fine blocks and bombs	cobbles
2-64	lapilli	gravel
0.0625-2	coarse ash	sand
<0.0625	fine ash	silt



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Pyroclasts from explosive eruptions

Tephra: the fragmental material produced by a volcanic eruption – regardless of composition, fragment size, or emplacement mechanism.



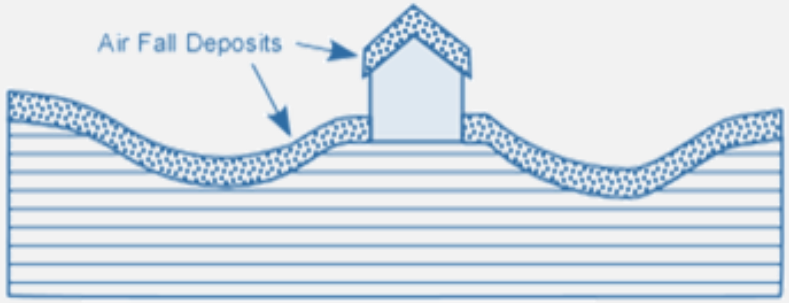
Tuff: a lithified rock made of tephra ejected by a volcanic eruption.

The components of a tuff are:

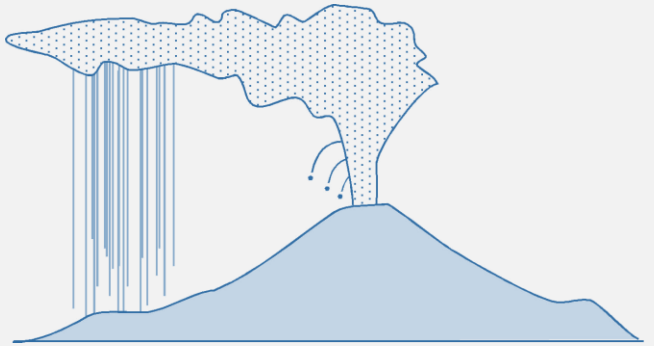
- Volcanic pumice, scoria and glass
- Crystals and crystal fragments
- Lithic (rock) fragments

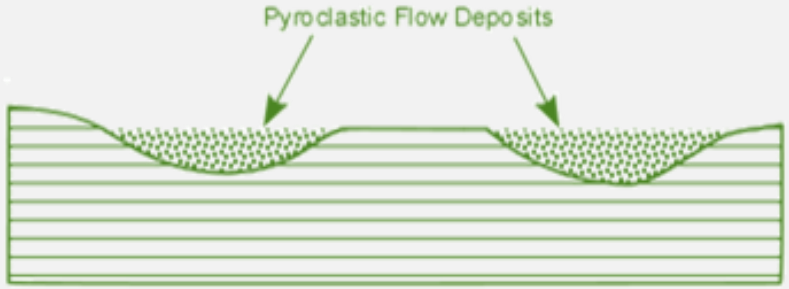
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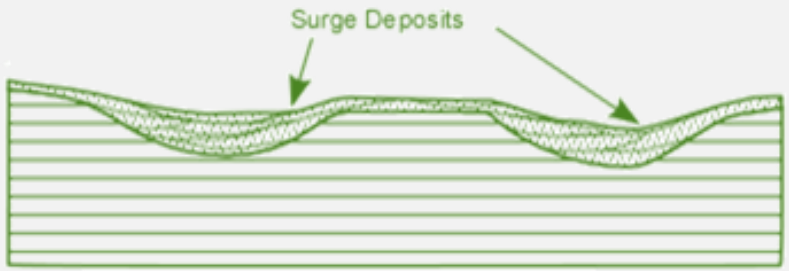
Pyroclastic deposits are subdivided into fall deposits and density current deposits



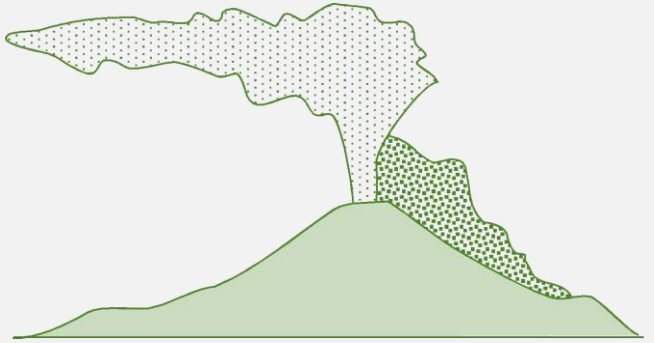
Pyroclastic fall deposits: fallout from a convecting eruption column







Pyroclastic density current deposits: eruption column collapse, or lava dome collapse



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Pyroclastic fall deposits from basaltic explosive eruptions

Hawaiian



Spatter cone

Strombolian



Scoria cone (cinder cone)

Low VEI means low plume height. Pyroclasts are deposited relatively close to the vent.

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Pyroclastic fall deposits from basaltic eruptions

Basaltic eruption products



Basaltic scoria



Pele’s hair



Pele’s tears (achneliths)

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Pyroclastic fall deposits from rhyolitic explosive eruptions

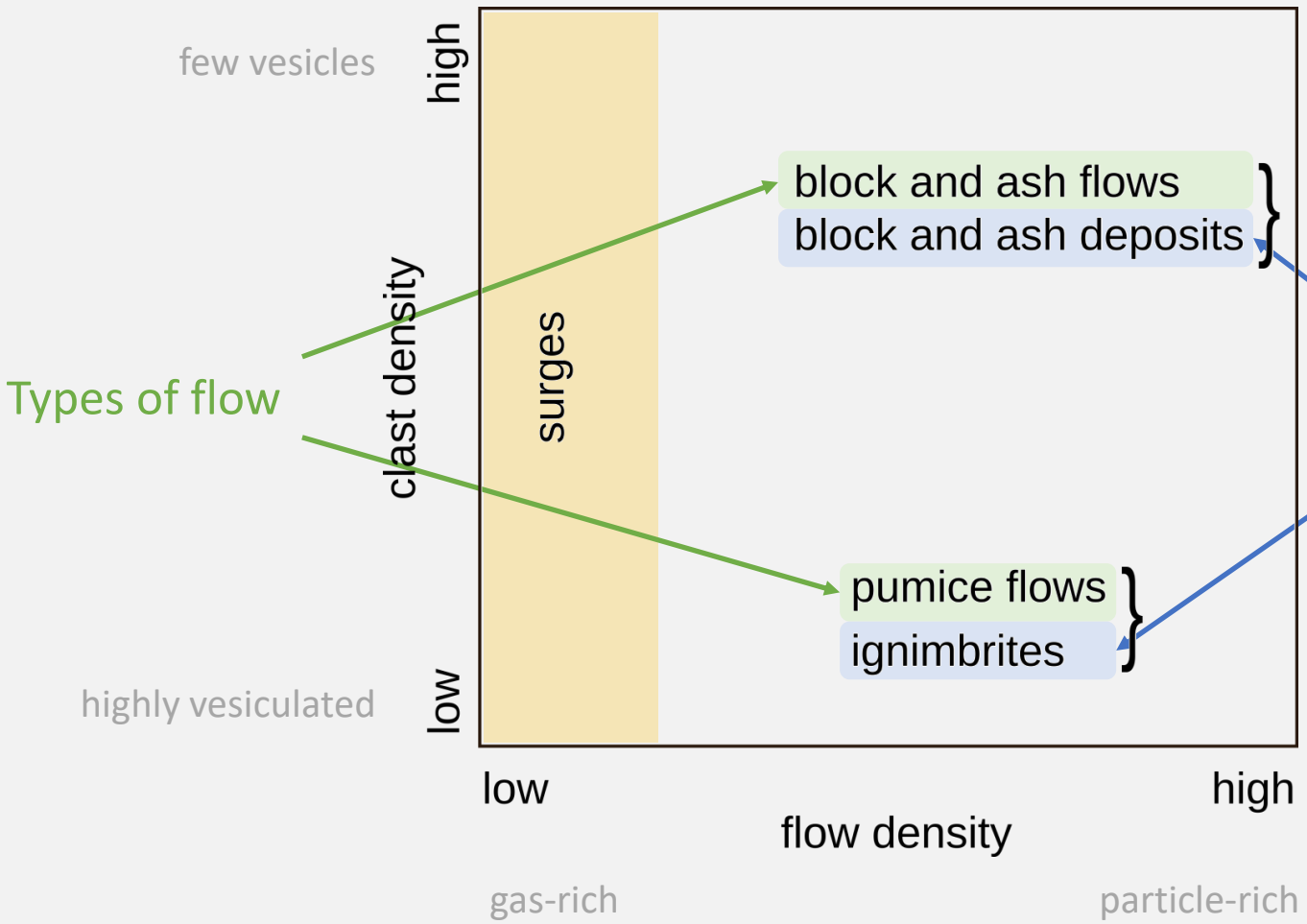
Plinian fall deposits: Chimborazo, Ecuador



Higher VEI means greater plume height. Pyroclasts are transported further from the vent; the smallest pyroclasts are transported furthest. Deposits are well sorted.

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Pyroclastic density currents



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Pyroclastic density currents are produced by column collapse and by dome collapse



Column collapse, Mayon Volcano, Philippines



Dome collapse, Soufrière Hills Volcano, Montserrat

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High-density PDCs with high-density clasts: Block-and-ash deposits







- Massive
- Very poorly sorted
- Dense lithic clasts in a fine matrix
- Lithics may coarsen upwards

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Low-density PDCs with low-density clasts: Pyroclastic surge deposits

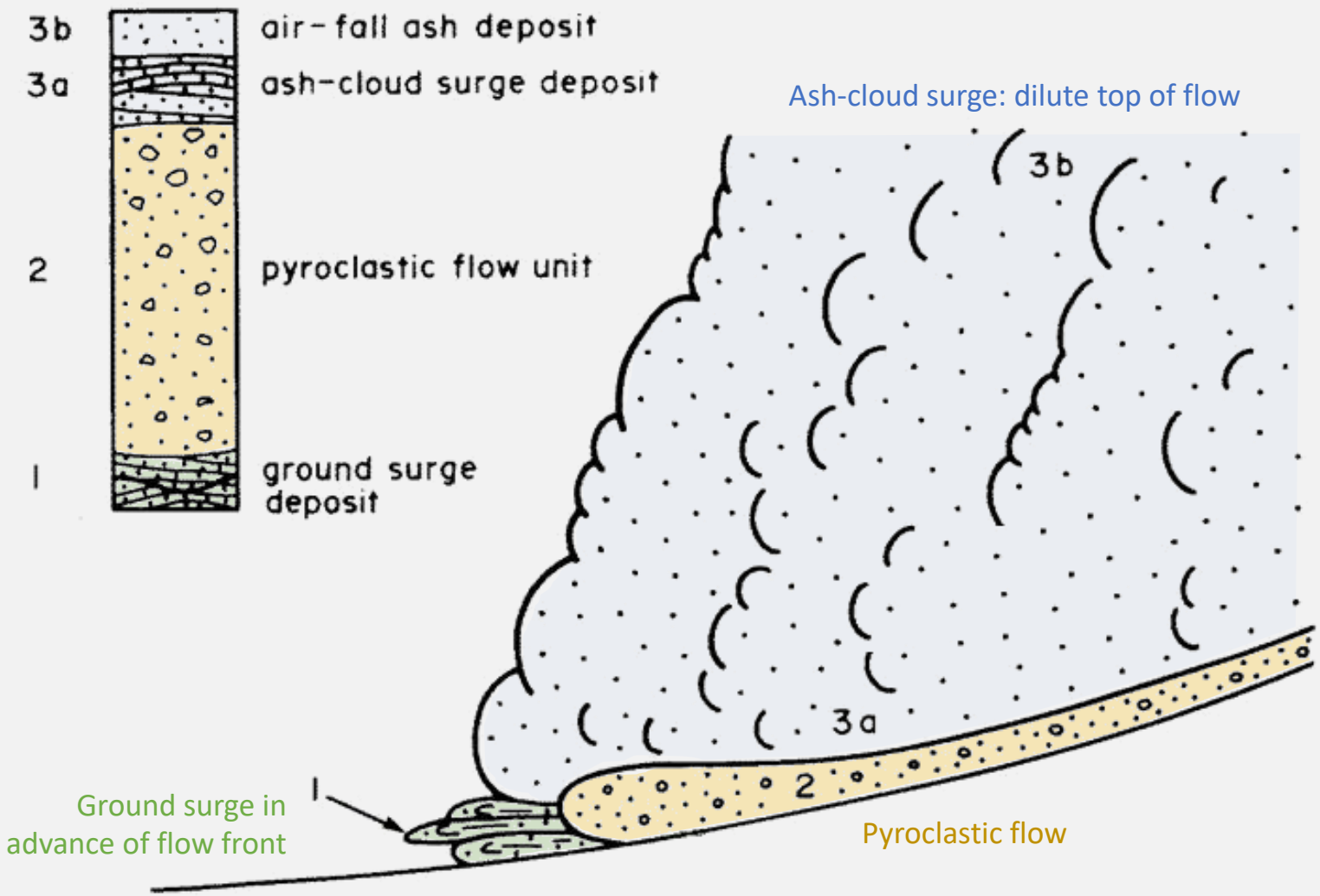




- Usually fine grained and well sorted
- Low-density juvenile clasts
- Prominent sedimentary structures

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High-density PDCs with low-density clasts: Ignimbrite deposits



3b: Fine ash fall*
3a: Ash-cloud surge*

2b: Flow
Reverse-graded pumice
Normally graded lithics

2a: Basal flow
Reverse-graded pumice
Reverse-graded lithics

1: Ground surge*

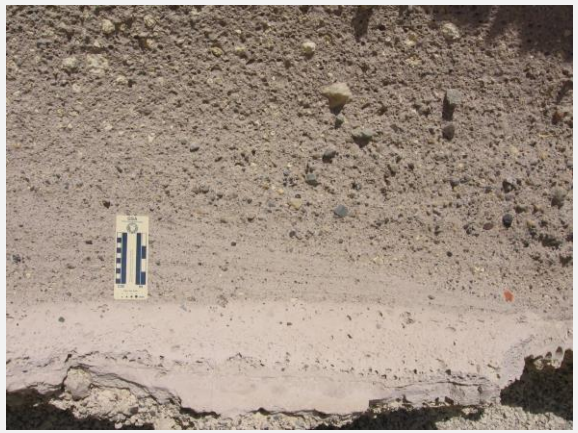
**not always present*

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Two types of ignimbrite deposit: unwelded and welded

Unwelded ignimbrites





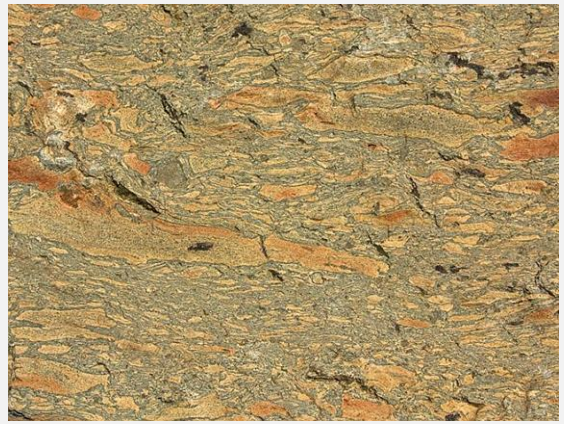


- Massive and poorly sorted: large pumice and lithic clasts in a fine ashy matrix
- May see reverse grading

Welded ignimbrites



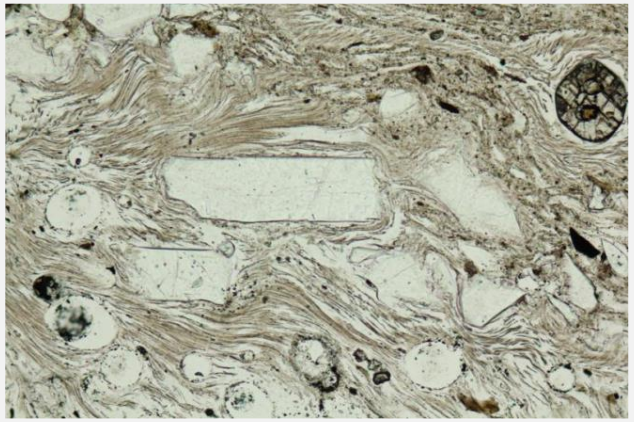




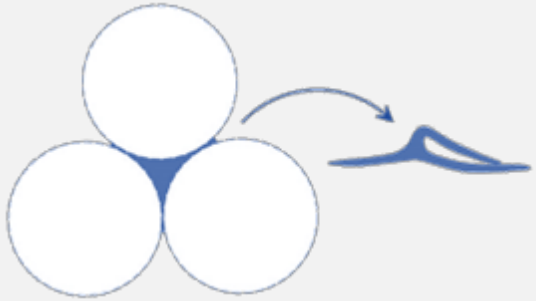
- Massive and poorly sorted
- Layered appearance due to collapsed, flattened and elongated pumices, referred to as fiammé
- Harder to see any original grading

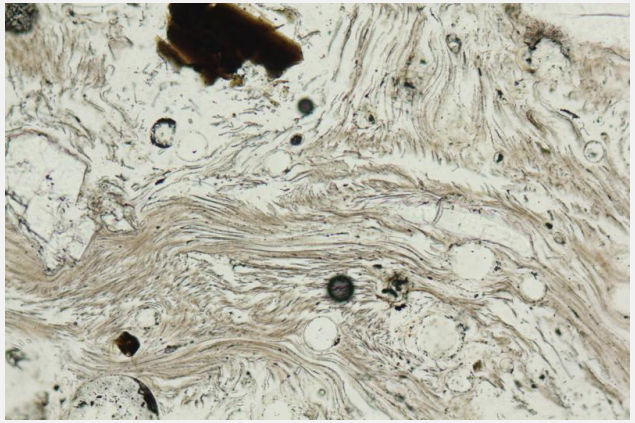
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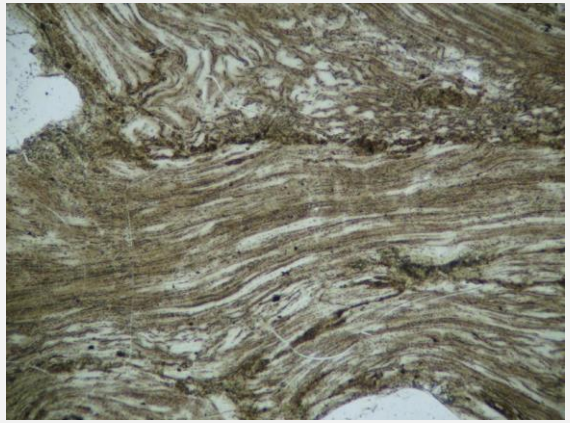
Eutaxitic texture in welded ignimbrites: What to look for in a thin section

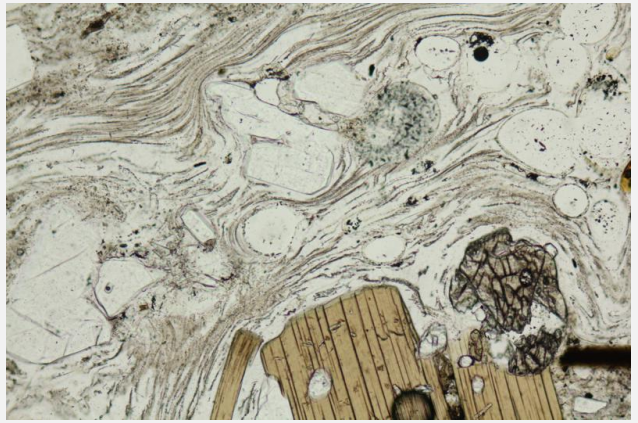


- A welded ignimbrite containing fiamme has a eutaxitic texture
- Layered or banded texture created by compaction of compaction of glass shards and pumice fragments
- Banding wraps around undeformed crystals and lithic fragments









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Volcanic eruptions

3.1 Explosive eruption styles and the Volcanic Explosivity Index

3.2 Products of explosive eruptions

- Pyroclastic fall deposits
- Pyroclastic density current deposits

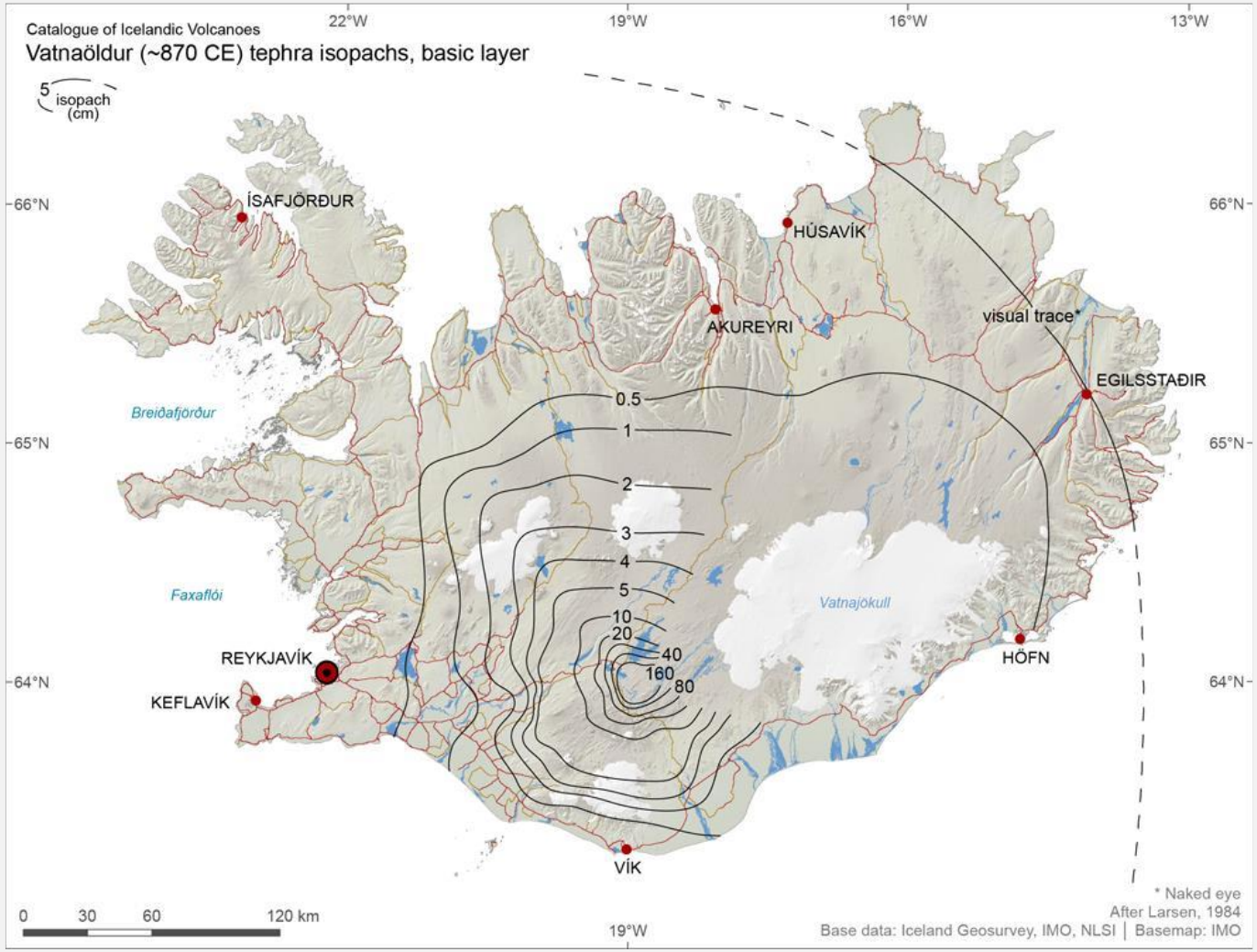
3.3 Reconstructing ancient eruptions from their deposits

3.4 Calderas



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Isopach maps: contours of equal tephra thickness



Isopach maps are used to calculate the total volume of a volcanic deposit

If the eruption volume is known, then the eruption magnitude can be determined using the VEI

Can you think of any limitations in this approach?

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Compound pyroclastic deposits at Santorini



Layer 1 (dark) was deposited in an older eruption

Layers 2-4 (light) were deposited during a younger eruption

Try interpreting these deposits yourself:

a) What types of deposit are shown here?

b) How did the eruption style change over the course of the younger eruption?

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Volcanic eruptions

You should now be able to:

- Classify tephra deposits according to their clast sizes and types
- Explain the volcanic processes that lead to the formation of pyroclastic fall deposits and pyroclastic density current deposits
- Interpret the volcanic processes that formed pyroclastic rocks using observations from outcrop to thin section scale



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Volcanic eruptions

3.1 Explosive eruption styles and the Volcanic Explosivity Index

3.2 Products of explosive eruptions

- Pyroclastic fall deposits
- Pyroclastic density current deposits

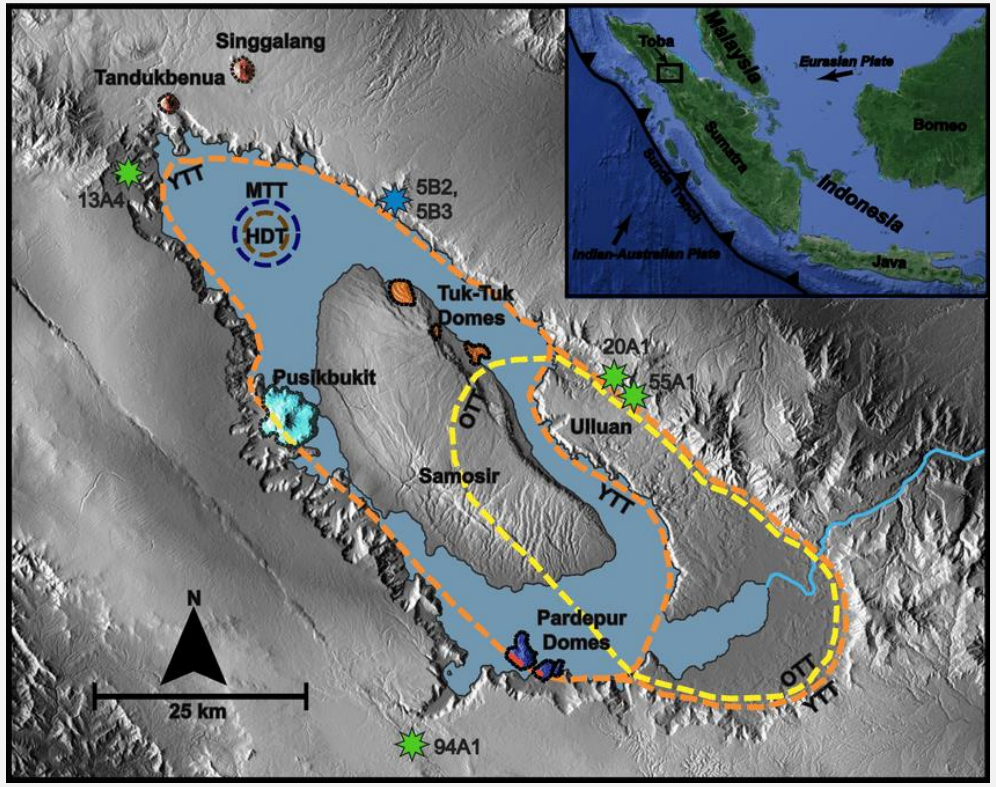
3.3 Reconstructing ancient eruptions from their deposits

3.4 Calderas



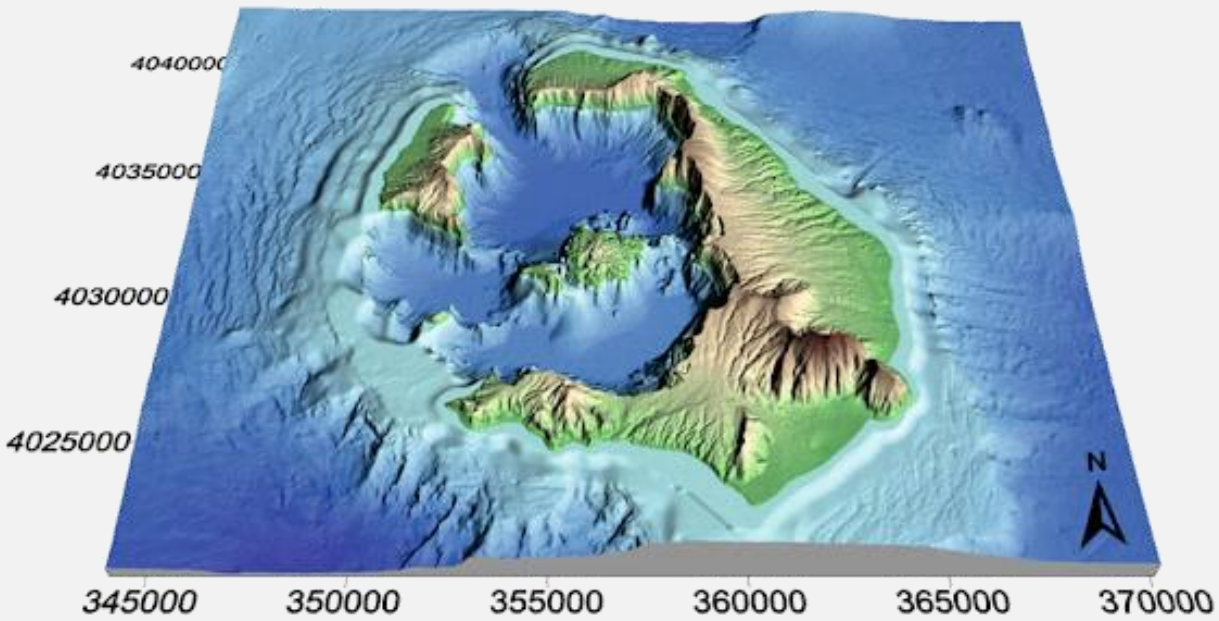
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The largest and most destructive eruptions leave calderas in the geological record



Toba caldera, Sumatra, Indonesia

Eruption ~74,000 years ago produced >3000 km^3 of magma, leaving a caldera 35 × 100 km



Santorini (Thera) caldera, Greece

Minoan Eruption ~3600 years ago produced ~40 km^3 of magma, leaving the modern caldera 8 × 11 km

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Calderas form when the roof of the magma chamber becomes structurally unsupported

Piston collapse

inward-dipping faults



outward-dipping faults



Trapdoor faulting



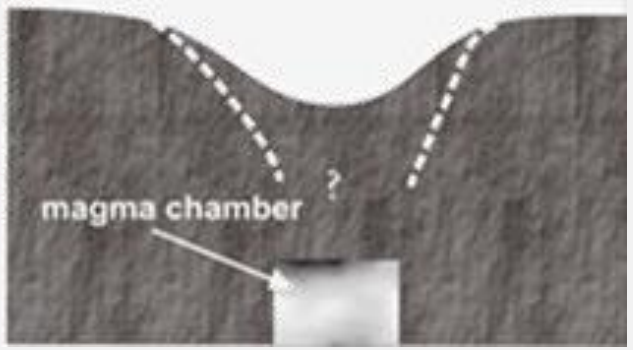
Piecemeal or chaotic collapse



Downsag

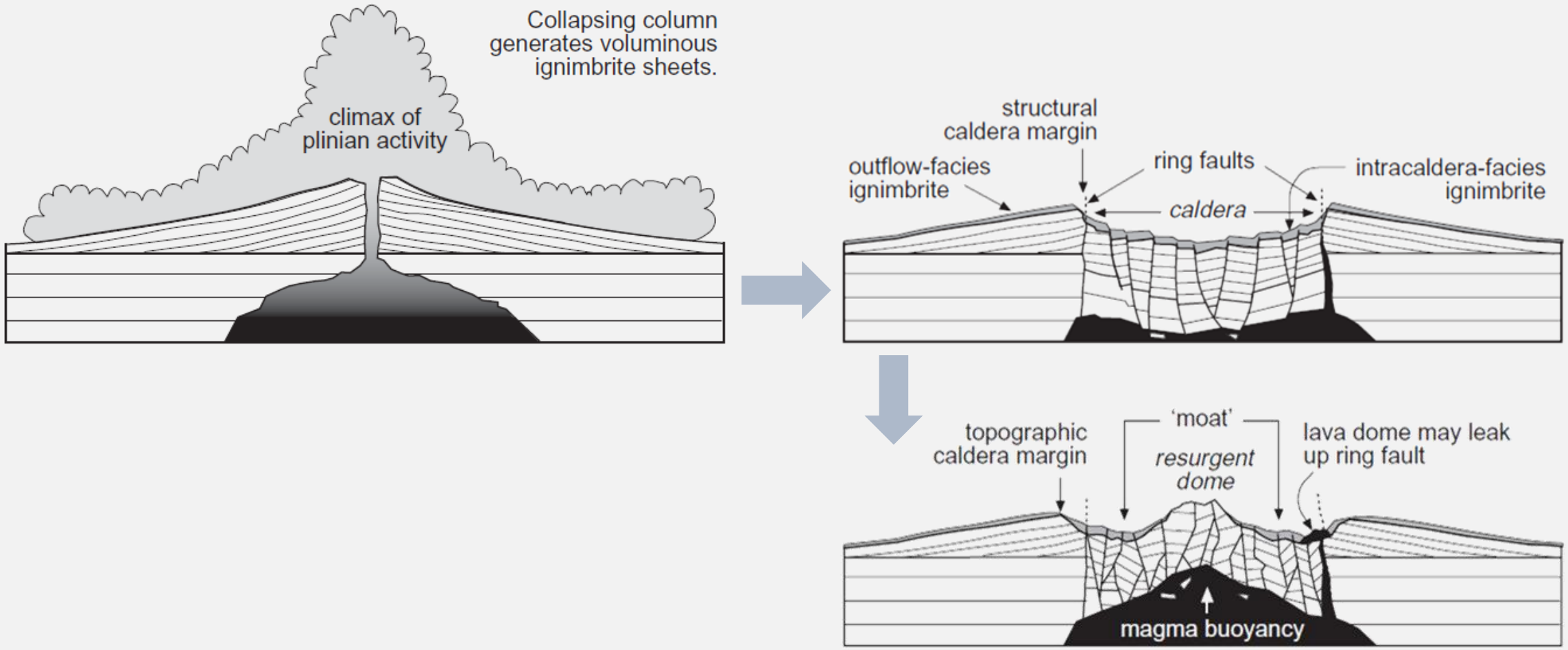


Funnel



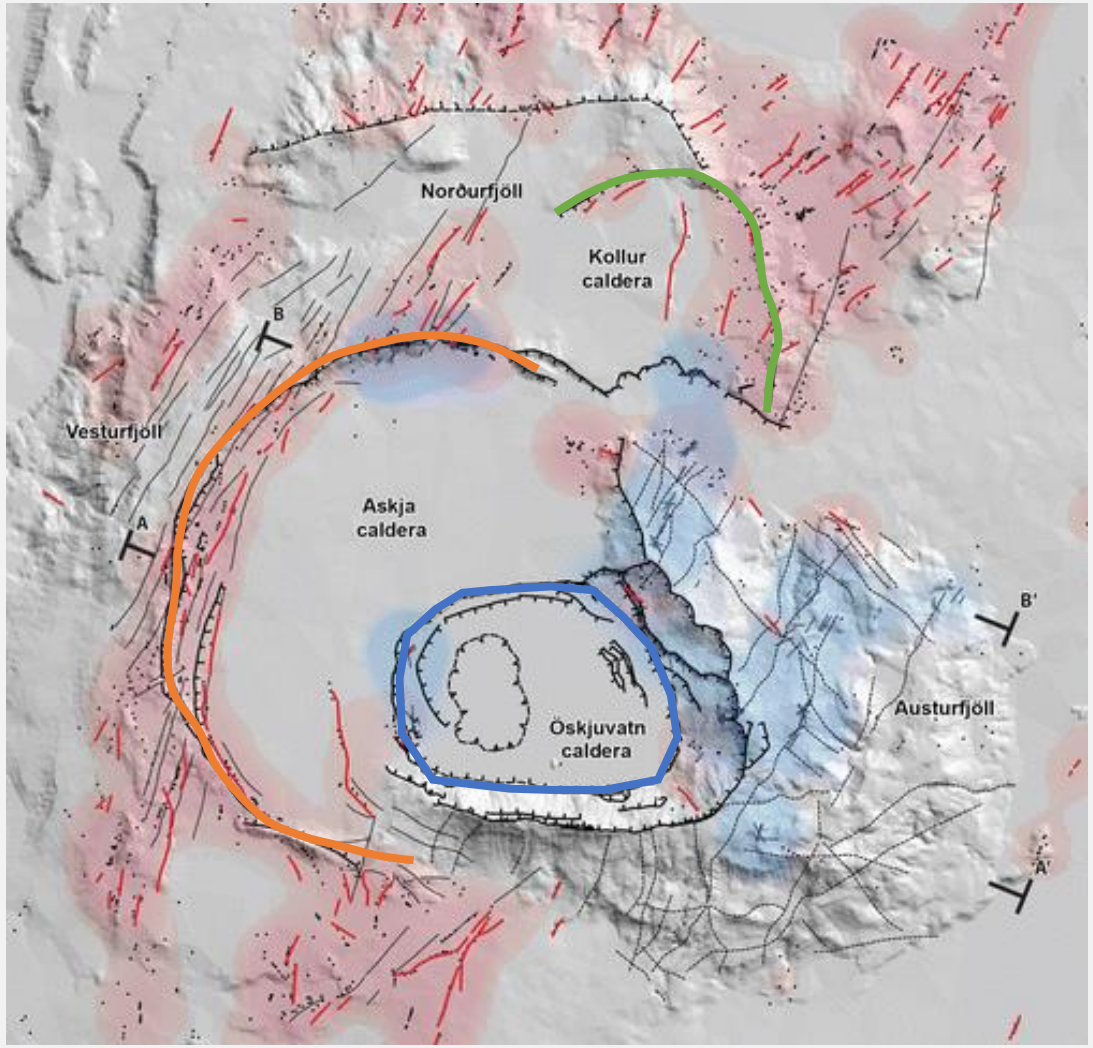
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Stages of formation of a complex caldera



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A present-day caldera volcano: Askja, North Iceland



Three nested calderas of different ages:

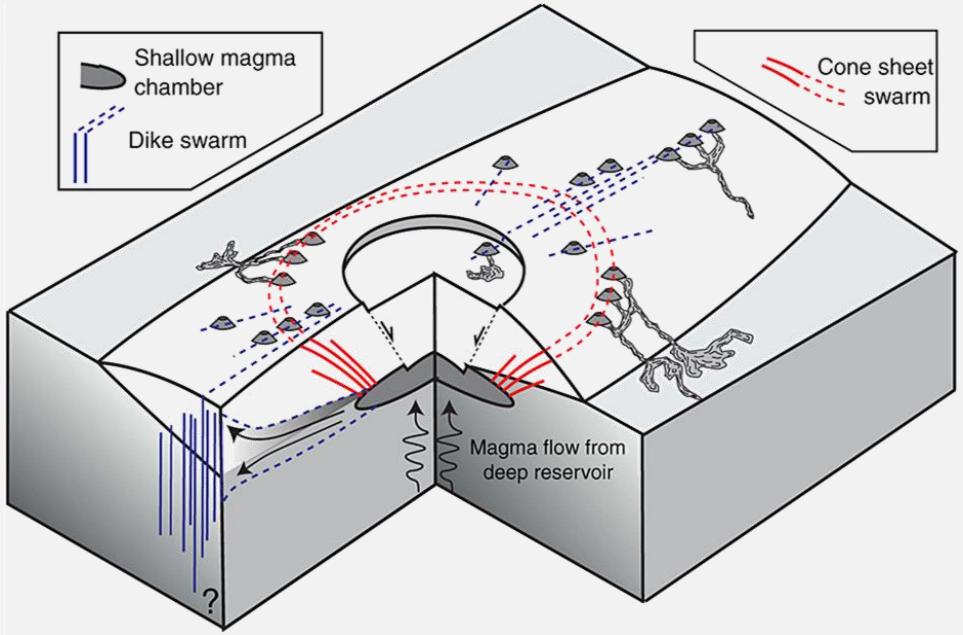
- Kollur: Oldest caldera; age uncertain
- Askja: Possibly formed about 10,000 years ago after a major rhyolitic eruption
- Öskjuvatn: formed after a Plinian eruption in 1875 CE. Now filled by a crater lake



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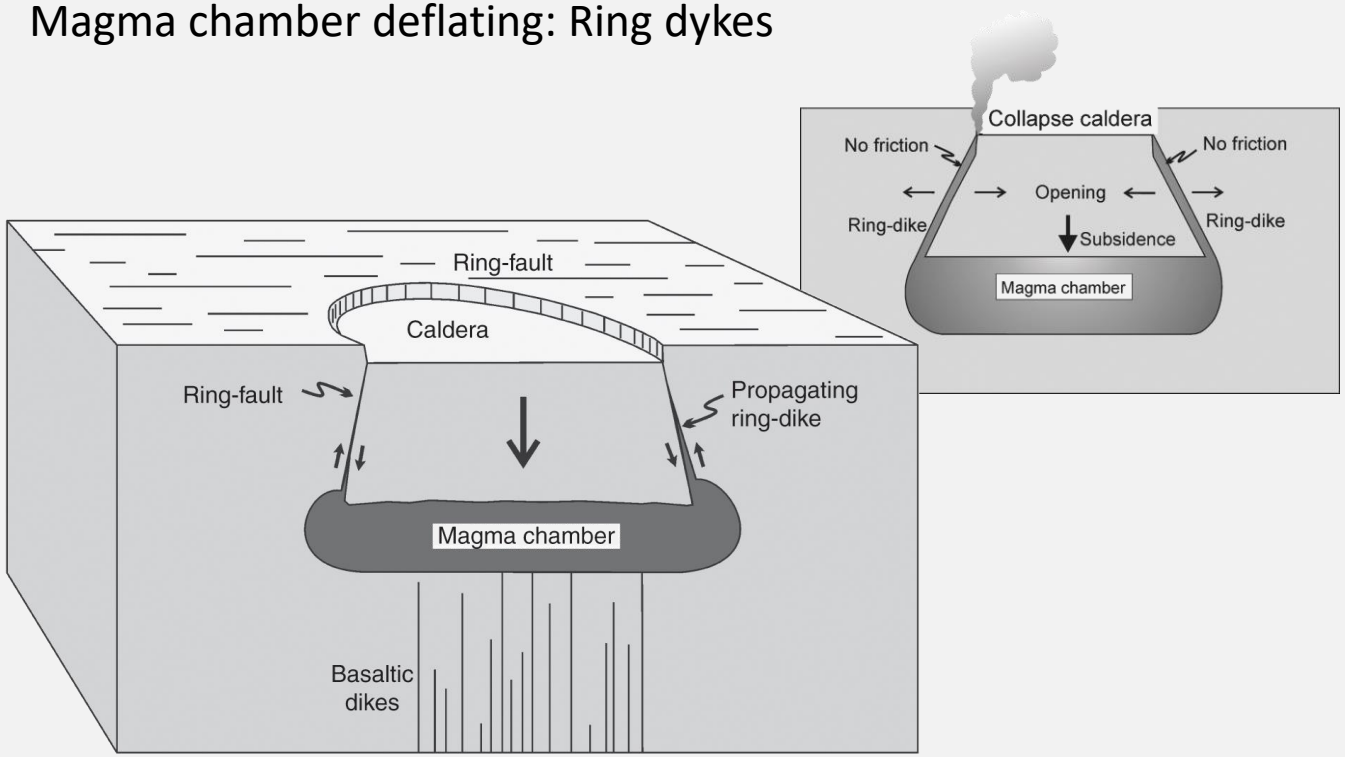
Cone sheets and ring dykes

Magma chamber inflating: Cone sheets



Dykes arranged in a curving array or set, that all dip steeply towards the magma chamber

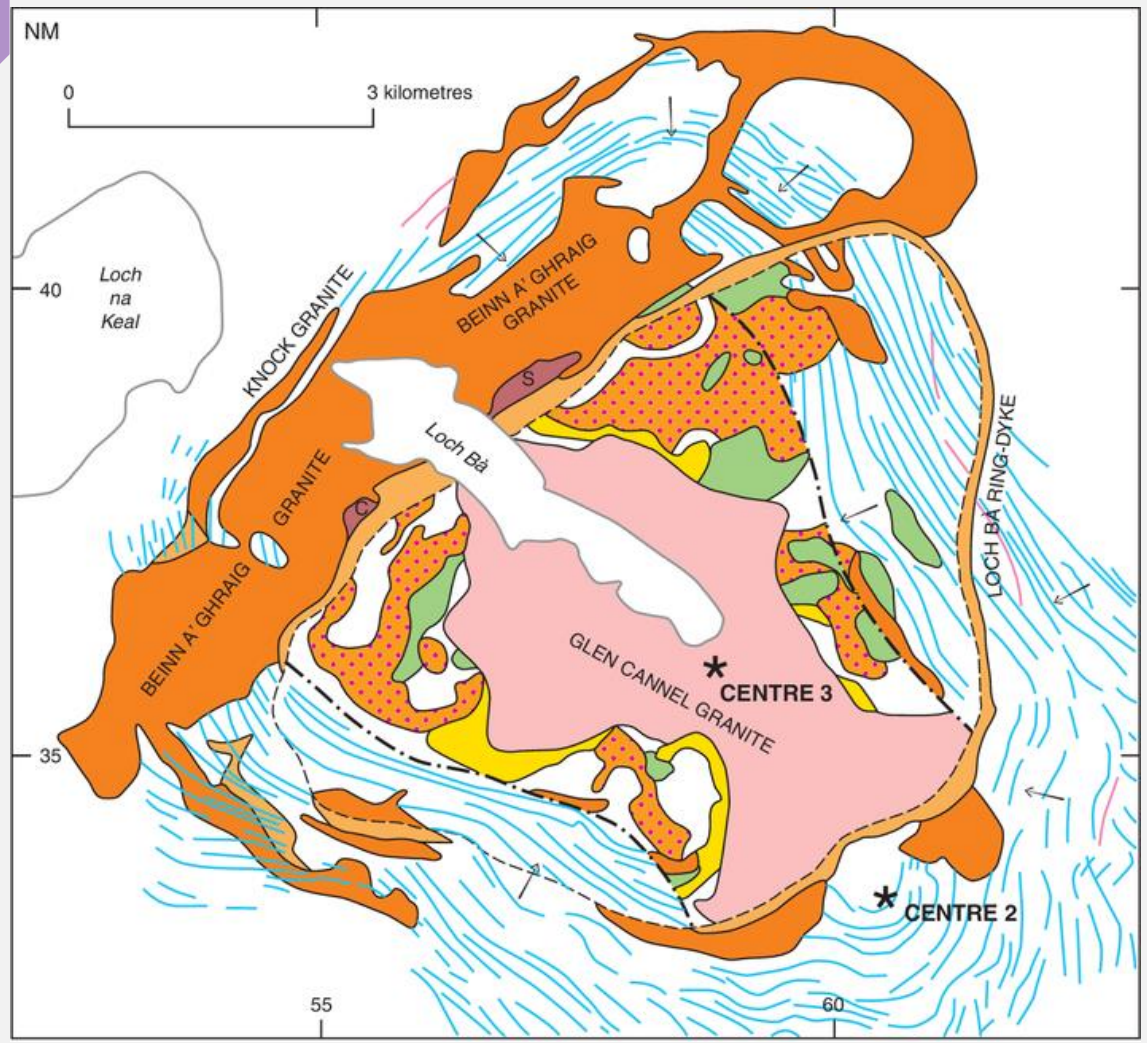
Magma chamber deflating: Ring dykes

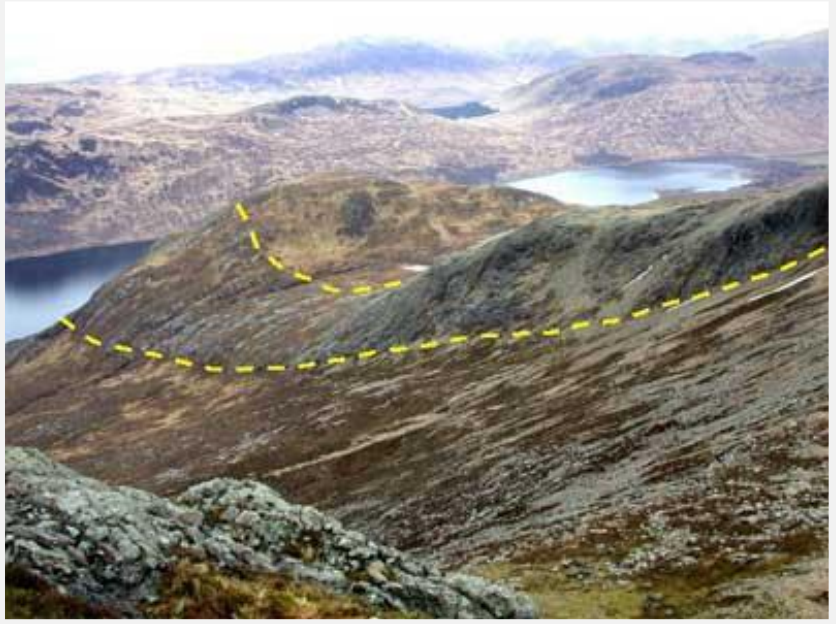


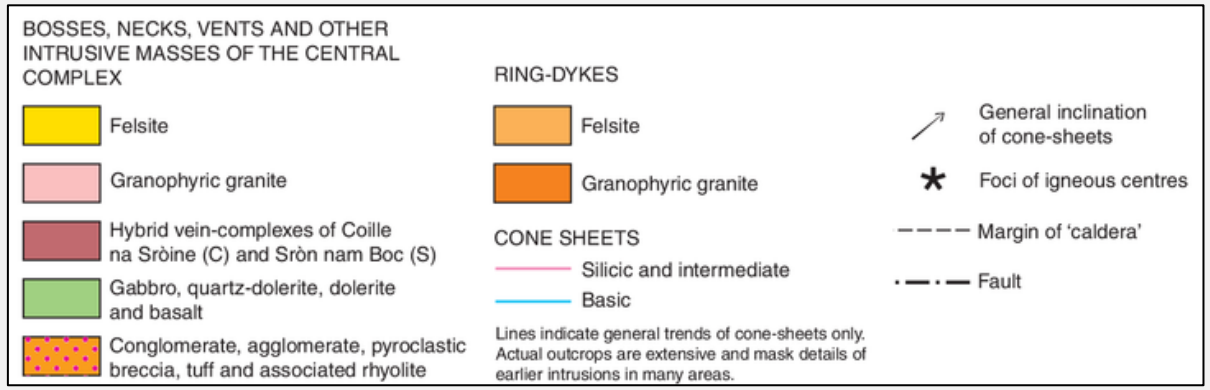
Intrusions that are circular or arcute in plan form, with steeply dipping contacts. Most commonly form after caldera collapse

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Cone sheets and ring dykes on the Isle of Mull, Scotland







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Volcanic eruptions

You should now be able to:

- Outline the processes that lead to caldera formation
- Describe how past caldera collapse events can be identified in the geological record



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